Homework — 1.2.1 Operating Systems — ANSWER SHEET

Separate answer sheet / mark scheme — keep for marking.

Mark scheme

Part A (14 marks)

1. [2] — 2 correct blanks per mark: (a) **abstraction**; (b) **platform**; (c) **multi-tasking**; (d) **scheduler**. *(Distractors not used: compression, defragmentation, interrupt.)*

2. [4] — 1 mark per cell:

Algorithm	How it decides	One drawback
FCFS	(a) Processes are executed in the order they arrive, each running until it completes	(b) One long process at the front of the queue makes every process behind it wait → poor average waiting time
SJF	(c) The process with the shortest total (estimated) run time is chosen next	(d) Long jobs may wait a very long time / indefinitely while shorter jobs keep being chosen; run times must be known or estimated in advance

3. [3] — 1 mark per row (correct verdict; for false statements the correction is required):

Statement	True / False	Correction
Scheduler time-slicing makes processes appear simultaneous	**True**	—
BIOS stored in RAM	**False**	The BIOS is stored in **ROM (flash)** because RAM is **volatile** and is empty when the computer is first switched on
SJF needs run times known in advance	**True**	—

4. [2] — Correct order:

Step	Order
Registers pushed onto the stack	3
CPU completes current FDE cycle and checks the interrupt register	1
Registers popped off the stack; original process resumes	5
Priority compared with current task	2
PC set to address of the ISR, which is executed	4

All five correctly placed = 2 marks; three or four correctly placed = 1 mark.

5. [3] — 1 mark: virtual memory uses an area of **secondary storage (page/swap file) as if it were RAM**. 1 mark: **pages** not currently needed are moved out of RAM to disk (and swapped back when needed), freeing space so the larger program can run. 1 mark: with too many large programs, pages are swapped between RAM and disk so frequently that the computer spends most of its time **transferring pages rather than doing useful work** — disk thrashing — so the machine slows to a crawl.

Part B (6 marks)

6. [2] — PC holds the address of the **next instruction** to be fetched (1). MAR holds the **address currently being accessed** in memory (the address copied from the PC so the instruction can be fetched) (1).

7. [2] — Multiple cores allow more than one instruction/thread to be executed truly in parallel, increasing throughput (1). It does not double performance because: not all tasks can be parallelised / some code is sequential; cores compete for shared resources (memory, cache, bus); overhead of coordinating/splitting work. Any one (1). Max 2.

8. [2] — SSD advantage (1): faster access/read-write, no moving parts so more durable/silent, lower power. HDD advantage (1): cheaper per GB / larger capacity for the price. Max 2.

Total: 14 + 6 = 20